Compact Objects Lab Report

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1 Introduction and Method

We used the StarShot network to run simulations on the evolution of compact objects in space. We explored how the object accretes mass from a companion star in a binary, and how its properties change as a function of column depth (depth from surface of object).

Most massive stars are part of binary systems. When one of the stars goes supernova, it will start to transfer its mass onto the other star. Figure 1 visualises this.

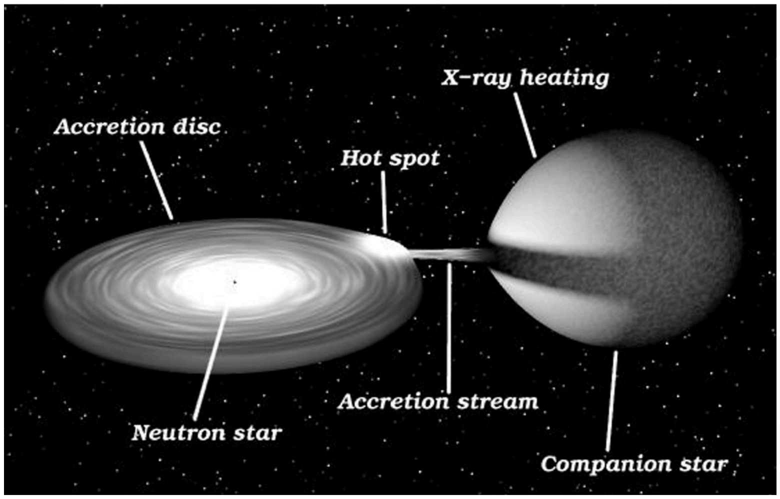

Figure 1: Diagram of an X-ray binary, where one star is supernova and is transferring its mass to the other star.

As the other star accretes mass, the mass undergoes thermonuclear burning. Once the temperature gets to a certain point, we see fusion reactions start to occur. As a result, we begin to see different heavier elements forming in the star.

During the in-class laboratory we used a companion star consisting of entirely He-4. This yielded some interesting results but was a very simple system designed to introduce us to the understanding of the content.

In this report we simulated the accretion of an object from a companion star consisting of purely carbon dioxide (CO_2). We looked at the material composition of the object, its luminosity, and its temperature. This was chosen as we believed it would produce some interesting results and we can explore how having two different elements will impact the star's evolution.

To reproduce these results, we followed the following steps.

Start with the initialisation steps in the StarShot virtual machine environment:

```
> ip
In [*]: %pylab
In [*]: ion()
```

Now we are in the Python environment.

```
In [*]: from starshot import Shot as S
s = S(Q=1.2, mdot=1, abu=dict(c12=12/44,o16=32/44), ymax=12)
```

We have included the fractional composition for CO_2, as well as specifying a simulation column depth of 10^{12} gcm^{-2}. The Q=1.2 parameter indicates that the star accretes 1.2 MeV of luminosity per nucleon. The luminosity is the Eddington luminosity: the critical luminosity where radiation pressure equals the force of gravity. It is given by

$$L_{\text{edd}} = \frac{4\pi G M m_{\text{p}} c}{\sigma_{\text{T}}}. \tag{1}$$

2 Results and Discussion

Figure 2 shows the change in percentage of composition in the compact object.

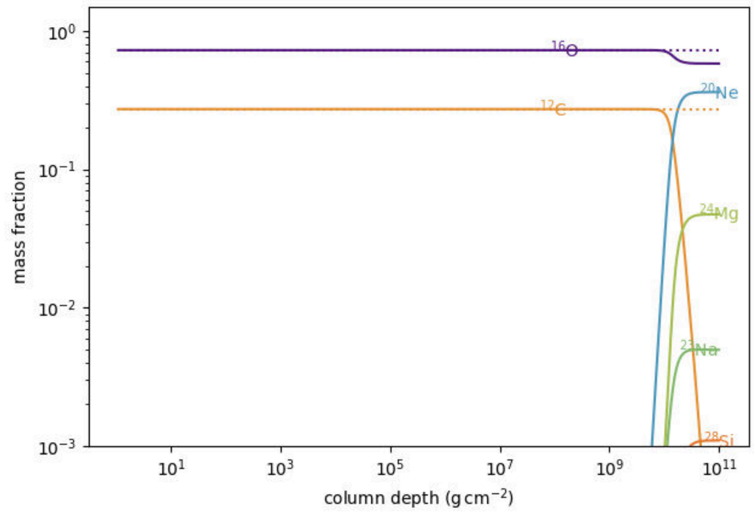

Figure 2: Composition of elements within stellar (compact) object as a function of column depth.

What we observe is that the object's composition remains constant for the majority of the depth of the object, and does not change from the initial distribution of elements. However, as the mass accretes, we start to see the high temperature of the object's surface causing nuclear fusion to occur, fusing our initial particles together to create many varying types of elements. This is an expected result, as the high pressure and temperature that arises from the extremely hot and dense accretion would cause much heavier elements to form via nuclear fusion.

An interesting result here is that only our carbon burns into other elements. The oxygen stays constant throughout our simulation, only burning partially into neon during the final stages. After the first nuclear burning, we have no more carbon but most of the oxygen remains. The reason for this is that oxygen has a higher energy level than carbon, and is a much more stable isotope. It is less likely to undergo nuclear fusion as it requires a higher temperature. Since carbon exists in abundance in our simulation, once the temperature is high enough for the fusion of carbon, the thermal energy turns into nuclear energy and does not continue to get high enough for much oxygen to fuse. It is entirely possible that if we continued the simulation, the oxygen would also begin to fuse into heavier elements.

We may also look at the Eddington luminosity of the object, in Figure 3.

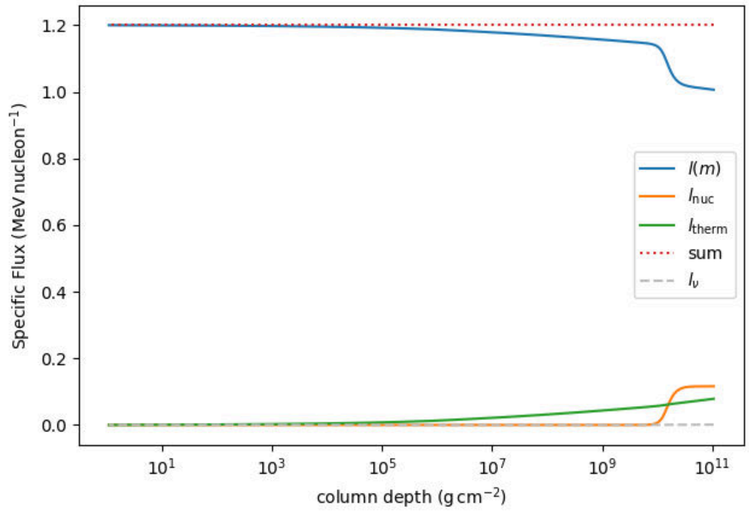

Figure 3: Luminosity as a function or column depth of compact CO2 object.

The dotted red line shows us that energy is conserved at every point in the compact object. Decreases in luminosity further inside the star must be compensated by rises in the nuclear and thermal energy in order to obey the conservation of energy. At the beginning of the plot, the object is emitting all its energy and is extremely luminous. Once the nuclear burning starts to occur, we see a rise in the thermal energy of the object, followed by a sharp rise in the nuclear energy once the carbon starts to burn. To obey the conservation of energy, the overall luminosity must decrease to facilitate the rise in thermal and nuclear energy. Interestingly, we see a notable sudden increase in the nuclear energy of the object that occurs at the same region where the carbon element burns to other elements (Neon, Magnesium, Sodium, and Silicon). From Equation 1, we see that luminosity is only a function of the mass of the accreted object. Evidently, once carbon starts to burn into other heavier elements, the mass accretes at a much faster rate, accelerating the effects of the nuclear energy. This also explains why we start to see the temperature rise in Figure 4 drop off after this point as more of the energy is used as nuclear energy than thermal energy.

We can also explore the temperature and density of the object as a function of column depth to see how the accretion and burning affects these properties (Figure 4).

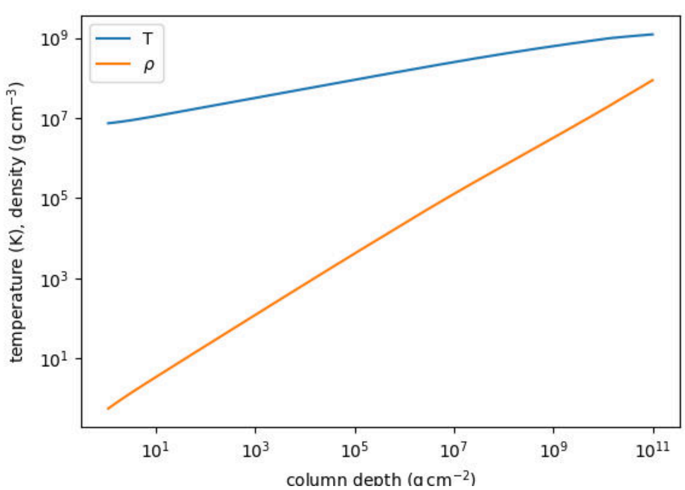

Figure 4: Temperature and density as a function of column depth.

The temperature of the accreted mass, as well as the density, both rise exponentially. The density rises much faster than the temperature. We can see a slight stagnation of the temperature rise towards the top end of the plot, notably around the same area where our carbon element burns into other elements. This is due to the fact that once our carbon begins to burn, the thermal energy within the star is transferred to nuclear energy. Since there is an overall slowing down in the increase of thermal energy, the temperature rise will also start to slow down.

3 Conclusion

Evidently, the accretion of mass in the creation of a neutron star is an interesting phenomenon. We explored how the different elements of the supernova star will impact the accretion of the neutron star, causing different elements in abundance. We also saw how the luminosity (and consequently, energy) changed throughout the accretion, as well as the density and temperature of the object. Some notable results we found was that the carbon will burn before the oxygen, and for the duration of our simulation the oxygen did not burn much, while the carbon burned entirely. We also saw how the thermonuclear reactions caused the creation of heavier elements, such as neon and magnesium, and eventually sodium and silicon. At the point of nuclear fusion, the luminosity of the star decreased as the energy was used as thermal and nuclear energy. We saw that the nuclear fusion resulted in the temperature increase slowing down, as more thermal energy was used up in the fusion reactions. In future it would be interesting to explore how other initial compositions might affect the results, as well as running the simulation for longer to see what kinds of elements we might end up with.